FlipBench: Measuring Directional Reasoning Asymmetry in Large Language Models

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Abstract

Current reasoning benchmarks evaluate large language models (LLMs) in a single direction—typically deriving conclusions from premises—conflating genuine understanding with pattern matching. We introduce FLIPBENCH, a benchmark of 1,200 matched forward-backward reasoning pairs across four domains (propositional logic, arithmetic, relational reasoning, and function computation) at three difficulty levels. Each problem instance has a forward variant (derive the conclusion) and a matched backward variant (recover the premise) testing the same underlying reasoning. All problems are programmatically generated with verified ground-truth answers. Evaluating five open-weight models (3.8B–32B parameters), we find that directional reasoning asymmetry is *domain-specific* rather than universally forward-biased: propositional logic and relational reasoning exhibit large forward biases (DRG up to 42.0 and 33.3 percentage points, respectively), while arithmetic and function computation sometimes show backward advantages. The asymmetry is stable across dataset seeds (std < 3 pp) and, in relational reasoning, amplifies significantly with problem difficulty ($p = 0.031$). Reasoning-optimized training (DeepSeek-R1) does not uniformly reduce the gap—it amplifies the logic DRG to 39.3 pp while eliminating it in function computation. These findings demonstrate that directional consistency is a diagnostic dimension complementary to aggregate accuracy, revealing qualitative differences in how models process forward versus backward inference.

1 Introduction

Large language models achieve impressive scores on reasoning benchmarks spanning logic [Parmar et al., 2024], mathematics, and commonsense inference. Yet growing evidence suggests that high benchmark accuracy may not reflect genuine, robust reasoning. The *Reversal Curse* [Berglund et al., 2024] demonstrated that LLMs trained on "A is B" fail to infer "B is A," revealing a fundamental directional asymmetry in factual knowledge retrieval. Subsequent work showed that autoregressive models exhibit systematic biases favoring forward reasoning over backward reasoning in planning tasks [Ren et al., 2025].

However, a critical gap remains: while individual papers have probed directional asymmetry in narrow settings—factual recall, planning, or isolated logic rules—**no systematic, cross-domain benchmark exists that uses matched forward-backward problem pairs to quantify directional reasoning asymmetry**. Without such a benchmark, we cannot answer fundamental questions: Is the asymmetry universal across reasoning types? Does it scale with problem difficulty? Do reasoning-optimized models reduce the gap?

We introduce FLIPBENCH, a benchmark designed to fill this gap. The core insight is that genuine reasoning should be *direction-invariant*: if a model truly understands a logical relationship, mathematical operation, or relational link, it should reason about it comparably in both the forward and backward directions. The gap between forward and backward performance on matched problem pairs—the **Directional Reasoning Gap** (DRG $=$ FA $-$ BA)—serves as a diagnostic signal separating genuine understanding from pattern matching.

Our contributions are:

- We introduce FLIPBENCH, a benchmark of 1,200 matched forward-backward reasoning pairs across four domains with three difficulty levels, all programmatically generated with verified ground-truth answers.

- We evaluate five open-weight models spanning 3.8B–32B parameters and two training paradigms (standard instruction tuning vs. reasoning-optimized RL training), revealing that directional asymmetry is *domain-specific* rather than universally forward-biased.
- We provide ablation studies on difficulty scaling, chain-of-thought prompting, and cross-seed stability, demonstrating that FLIPBENCH captures a stable, meaningful diagnostic dimension complementary to aggregate accuracy.

The remainder of the paper is organized as follows. Section 2 reviews related work. Section 3 describes the FLIPBENCH benchmark design. Section 4 presents experimental results including main comparisons, ablations, and error analysis. Section 5 discusses findings and limitations. Section 6 concludes.

2 Related Work

The Reversal Curse. Berglund et al. [2024] showed that autoregressive LLMs trained on "A is B" cannot infer "B is A," demonstrating a fundamental asymmetry in factual knowledge storage. Follow-up work by Golovneva et al. [2024] analyzed and proposed mitigations, while Allen-Zhu and Li [2025] connected the phenomenon to transformer binding limitations. Our work extends this analysis from factual recall to *reasoning*: we test whether the directional asymmetry that plagues knowledge retrieval also affects logical inference, mathematical computation, and relational reasoning.

Logical Reasoning Benchmarks. LogicBench [Parmar et al., 2024] systematically evaluates 25 logical reasoning patterns but does not create matched pairs to measure the directional gap on the same underlying problem. SATBench [Wei et al., 2025] generates logic puzzles from SAT formulas, and ZebraLogic [Lin et al., 2025] tests constraint satisfaction at varying complexity. Our contribution is orthogonal: we focus on directional consistency rather than overall reasoning ability.

Forward vs. Backward Reasoning in LLMs. Ren et al. [2025] found that LLMs struggle with backward planning due to autoregressive generation biases. Surveys on empowering LLMs with logical reasoning [Wang et al., 2025a] note this forward bias but provide no systematic benchmark. Our work fills this gap with a dedicated benchmark resource.

Benchmark Quality and Meta-Evaluation. Benchmark^2 [Qian et al., 2026] proposed meta-evaluation metrics for LLM benchmarks. Item Response Theory approaches [Zhou et al., 2026] diagnose benchmark item quality. Benchmark Profiling [Kim et al., 2025] uses mechanistic interpretability to identify which abilities benchmarks test. Our work introduces a new diagnostic dimension—directional reasoning consistency—that these approaches do not capture.

Reasoning Robustness. ReasonBENCH [Wang et al., 2025b] measures stochastic instability across multiple runs. FLIPBENCH is complementary: we test *structural invariance* (same reasoning, reversed direction) rather than surface-level perturbations. Similarly, the generation-verification gap literature [Yan et al., 2025] studies asymmetry between generating and verifying solutions, whereas we compare two forms of generation (forward and backward).

3 Method: The FLIPBENCH Benchmark

3.1 Overview

FLIPBENCH consists of 1,200 matched forward-backward reasoning pairs across four domains, each with three difficulty levels (100 pairs per level per domain). Every problem instance has a **forward variant** and a matched **backward variant** that test the same underlying reasoning operation in opposite directions. All problems are programmatically generated with verified ground-truth answers, enabling contamination-resistant evaluation and arbitrary scaling.

3.2 Domain Design

Domain 1: Propositional Logic (300 pairs). Forward problems present a set of propositional rules (e.g., "If P then Q") and initial facts, asking the model to derive whether a target proposition is true or false via forward chaining. Backward problems present the same rules and the target proposition's truth value, asking the model to determine what initial facts must hold via backward chaining.

- **Difficulty 1**: 1-hop (single rule application)
- **Difficulty 2**: 2-hop (chained rules)
- **Difficulty 3**: 3-hop (multi-step chaining with distractor rules)

Domain 2: Arithmetic Reasoning (300 pairs). Forward problems provide a mathematical expression or word problem with all operands and ask the model to compute the result. Backward problems provide the result and all-but-one operand, asking the model to find the missing value.

- **Difficulty 1**: Single operation
- **Difficulty 2**: Two chained operations
- **Difficulty 3**: Three chained operations with mixed operators

Domain 3: Relational Reasoning (300 pairs). Forward problems present entity-relation facts (family relationships) and ask the model to derive a target relation by composing given relations. Backward problems present a compound relationship and ask the model to decompose it—identifying what base relationship holds.

- **Difficulty 1**: Single-hop relations
- **Difficulty 2**: Two-hop transitive relations
- **Difficulty 3**: Three-hop with distractor entities

Domain 4: Function Computation (300 pairs). Forward problems present a simple Python function and an input value, asking the model to compute the output. Backward problems present the same function and a target output, asking for the input that produces it. Functions are restricted to be invertible on the integer domain $[1, 50]$.

- **Difficulty 1**: Single-operation functions (e.g., $f(x) = ax + b$)
- **Difficulty 2**: Two composed operations
- **Difficulty 3**: Three composed operations

3.3 Design Principles

Matched difficulty. Forward and backward variants of the same pair involve identical reasoning depth. The backward variant reverses the direction of inference, not its complexity.

Verified ground truth. All instances are programmatically generated with deterministic, verifiable answers. No human annotation is needed, eliminating label noise.

Controlled generation. Each domain uses templated generation with randomized parameters from specified random seeds, enabling reproducibility and minimizing memorization risk.

Multiple seeds. We generate three independent dataset versions (seeds 42, 123, 456) to assess cross-seed stability.

3.4 Metrics

We define the following metrics:

- **Forward Accuracy (FA)**: Accuracy on forward variants
- **Backward Accuracy (BA)**: Accuracy on backward variants
- **Directional Reasoning Gap (DRG)**: FA − BA. Positive values indicate forward bias; negative values indicate backward advantage.
- **Consistency Rate ()**: Fraction of pairs where the model answers *both* forward and backward correctly. If $< \min(\text{FA}, \text{BA})$, the model fails on *different* instances in each direction, indicating direction-dependent error patterns rather than a uniform accuracy offset.

3.5 Evaluation Protocol

Models are evaluated zero-shot with a standardized prompt template that is direction-agnostic: a system message ("You are a precise reasoning assistant. Answer the question below. Give your final answer on the last line in the format: ANSWER: <your answer>") followed by the problem text. Temperature is set to 0 for deterministic outputs. Answers are parsed from the model output by extracting text after "ANSWER:" with domain-specific parsing (True/False for logic, integers for arithmetic and functions, entity names for relational).

4 Experiments

4.1 Setup

Models. We evaluate five open-weight models spanning different sizes and training paradigms:

1. **Phi-3.5-mini** (3.8B) — small general-purpose model [Abdin et al., 2024]
2. **Llama-3.1-8B-Instruct** — medium-size general-purpose [Grattafiori et al., 2024]
3. **Qwen2.5-7B-Instruct** — medium alternative family [Yang et al., 2024]
4. **DeepSeek-R1-Distill-Qwen-7B** — reasoning-optimized via RL training [Guo et al., 2025]
5. **Qwen2.5-32B-Instruct-AWQ** — larger model (4-bit quantized) [Yang et al., 2024]

This selection covers multiple model families, size scaling (3.8B → 32B), and standard versus reasoning-optimized training. All models are run using vLLM [Kwon et al., 2023] with float16 precision on a single NVIDIA RTX A6000 (48GB).

Table 1: Main FLIPBENCH results (seed 42). FA = Forward Accuracy (%), BA = Backward Accuracy (%), DRG = FA $-$ BA (pp), CR = Consistency Rate (%). Bold indicates the largest |DRG| per domain. † denotes statistical significance at $p < 0.05$ (McNemar's test with Bonferroni correction).

	Propositional Logic				Arithmetic				Relational				Function			
Model	FA	BA	DRG	CR	FA	BA	DRG	CR	FA	BA	DRG	CR	FA	BA	DRG	CR
Phi-3.5-mini (3.8B)	86.3	85.7	0.7	73.0	71.0	54.7	16.3†	50.0	78.3	45.0	**33.3**†	31.7	39.0	93.0	**−54.0**	37.7
Llama-3.1-8B	61.3	50.0	11.3	23.3	84.7	98.7	−14.0	83.7	77.7	53.0	24.7†	43.3	87.0	99.7	−12.7	86.7
Qwen2.5-7B	85.0	67.7	17.3†	53.0	88.3	83.3	5.0	78.3	67.3	43.7	23.7†	30.3	100.0	99.3	0.7	99.3
DeepSeek-R1-7B	88.0	48.7	39.3†	39.0	56.7	75.7	−19.0	55.7	50.0	29.0	21.0†	17.3	99.3	99.7	−0.3	99.0
Qwen2.5-32B	100.0	58.0	**42.0**†	58.0	100.0	98.0	2.0	98.0	85.0	67.0	18.0†	56.0	100.0	100.0	0.0	100.0

Baselines. We compute random-guessing baselines: 50% for propositional logic (binary True/False), ≈0% for arithmetic and function computation (open-ended integers), and $1/N$ for relational reasoning (choosing from N mentioned entities).

Seeds. All models are evaluated on seed 42. Llama-3.1-8B and DeepSeek-R1-7B are additionally evaluated on seeds 123 and 456 for cross-seed stability analysis.

4.2 Main Results

Table 1 presents the main results across all models and domains.

Several key patterns emerge:

Directional asymmetry is domain-specific, not universally forward-biased. Contrary to our initial hypothesis that forward reasoning would consistently outperform backward reasoning, the direction of asymmetry depends strongly on the domain. Propositional logic and relational reasoning show consistent forward biases across all models (DRG 0.7–42.0 pp for logic; 18.0–33.3 pp for relational). In contrast, arithmetic and function computation often exhibit *backward advantages*: Llama-3.1-8B achieves 98.7% backward accuracy versus 84.7% forward on arithmetic (DRG $= -14.0$), and Phi-3.5-mini shows a striking -54.0 pp DRG on function computation.

Relational reasoning shows the most consistent forward bias. All five models exhibit statistically significant forward bias in relational reasoning (DRG 18.0–33.3 pp, all $p < 0.05$ after Bonferroni correction), making this the domain with the most reliable directional asymmetry.

Propositional logic shows the largest absolute DRG magnitudes. Qwen2.5-32B achieves perfect 100% forward accuracy on propositional logic but only 58.0% backward accuracy (DRG $= 42.0$ pp). DeepSeek-R1-7B shows a similar pattern (88.0% vs. 48.7%, DRG $= 39.3$ pp).

Function computation is largely direction-invariant for capable models. Qwen2.5-7B, DeepSeek-R1-7B, and Qwen2.5-32B all achieve near-perfect accuracy in both directions (DRG ≤ 0.7 pp), suggesting these models have genuinely learned function evaluation and inversion for simple functions.

4.3 Consistency Analysis

The consistency rate reveals that models often fail on *different* instances depending on direction. Table 1 shows that CR is consistently below $\min(\text{FA}, \text{BA})$ in most model-domain combinations. For example, Llama-3.1-8B in propositional logic has FA $= 61.3\%$ and BA $= 50.0\%$, but CR $= 23.3\%$—far below the 50.0% that would be expected if backward errors were simply a subset of forward errors. This confirms that the DRG reflects genuine *directional* variation, not merely a uniform accuracy offset.

Figure 1 breaks down pair-level outcomes into four categories: both correct, forward-only correct, backward-only correct, and both wrong.

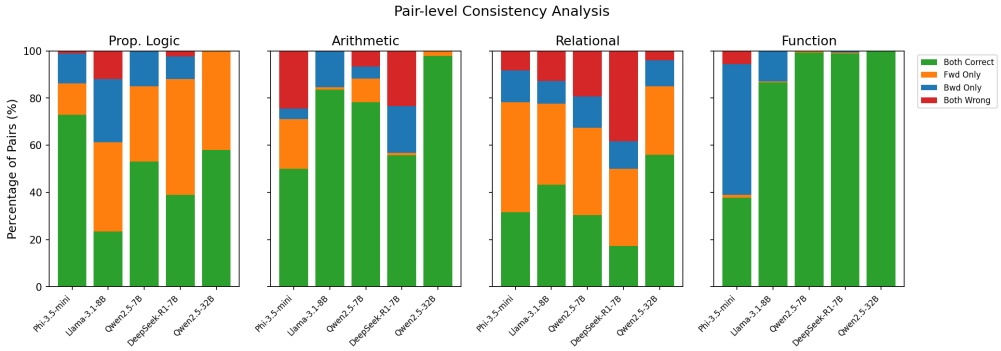


Figure 1: Pair-level outcome breakdown per model and domain. "Fwd only" and "Bwd only" segments indicate instances where the model succeeds in one direction but fails in the other. Large asymmetric segments (e.g., DeepSeek-R1-7B in logic) confirm direction-dependent error patterns.

4.4 Ablation 1: DRG Across Difficulty Levels

Figure 2 shows how DRG varies with problem difficulty across the four domains.

Relational reasoning shows clear difficulty amplification. The mean DRG across models increases from -15.8 pp at difficulty 1 to $+54.8$ pp at difficulty 3 (Wilcoxon signed-rank test $p = 0.031$). This is the only domain where the DRG monotonically increases with difficulty in a statistically significant manner.

Propositional logic shows modest amplification. Mean DRG increases from 19.2 pp (easy) to 24.4 pp (hard), with an amplification factor of 1.27, though this is not statistically significant ($p = 0.44$).

Arithmetic and function domains show non-monotonic patterns. In arithmetic, several models (Llama-3.1-8B, DeepSeek-R1-7B) show a *backward* advantage that grows with difficulty—harder forward problems suffer more than harder backward problems. In function computation, Phi-3.5-mini's backward advantage diminishes with difficulty (DRG from -85 pp at difficulty 1 to -22 pp at difficulty 3).

4.5 Ablation 2: Standard vs. Reasoning-Optimized Training

We compare Qwen2.5-7B-Instruct (standard instruction tuning) with DeepSeek-R1-Distill-Qwen-7B (reasoning-optimized via RL), both based on similar architectures at the 7B scale. Figure 3 shows the comparison.

Reasoning-optimized training does not uniformly reduce directional asymmetry. DeepSeek-R1-7B has a *larger* DRG than Qwen2.5-7B in propositional logic (39.3 vs. 17.3 pp) but a *smaller* DRG in relational reasoning (21.0 vs. 23.7 pp) and essentially zero DRG in function computation (-0.3 vs. 0.7 pp). In arithmetic, both models show opposite-sign DRGs (Qwen: $+5.0$; DeepSeek-R1: -19.0), reflecting different failure modes rather than a simple reduction.

4.6 Ablation 3: Chain-of-Thought Prompting

We evaluate whether adding "Think step by step before giving your final answer" to the prompt reduces the DRG. Table 2 shows results for Llama-3.1-8B and DeepSeek-R1-7B with and without chain-of-thought (CoT) prompting.

CoT prompting has mixed effects. For Llama-3.1-8B, CoT reduces the logic DRG from 11.3 to 2.0 pp and the arithmetic DRG from -14.0 to 0.3 pp, but barely affects the relational DRG (24.7 $\rightarrow$ 23.0 pp). For DeepSeek-R1-7B, CoT slightly *increases* the logic DRG (39.3 $\rightarrow$ 48.3 pp) while reducing

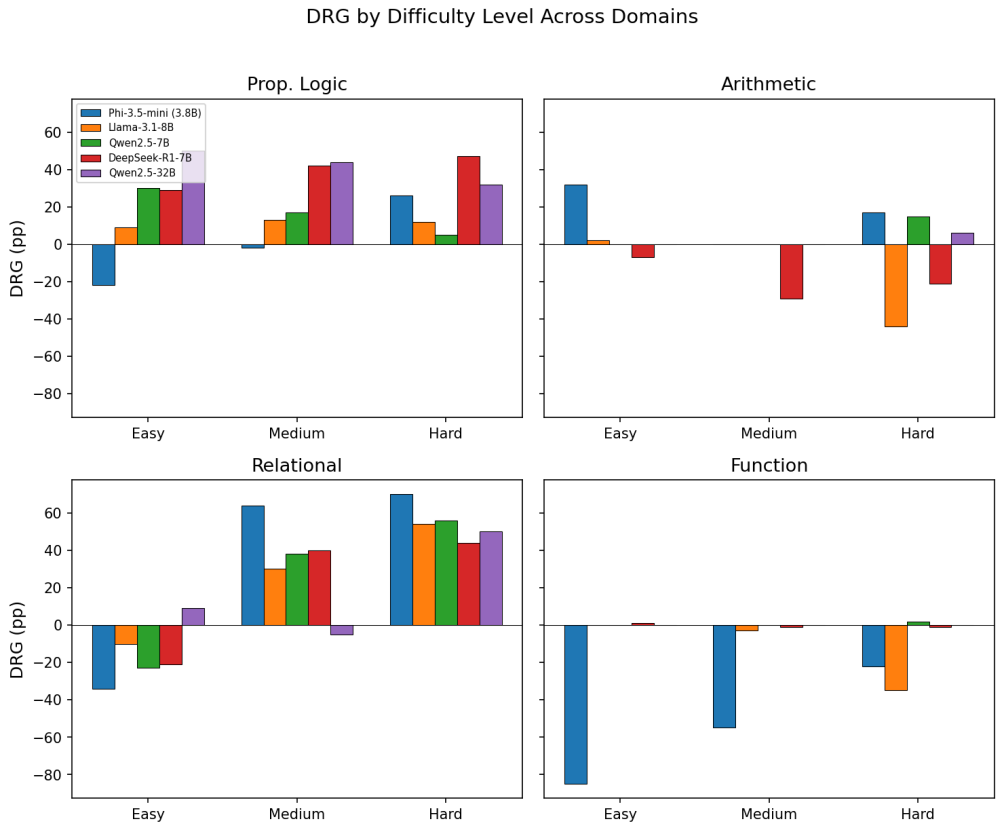


Figure 2: DRG across difficulty levels by domain (seed 42). In relational reasoning, the DRG increases significantly from difficulty 1 to difficulty 3 ($p = 0.031$, Wilcoxon signed-rank test). Other domains show more complex patterns.

the arithmetic DRG ($-19.0 \rightarrow -1.7$ pp). This suggests that CoT helps in domains where the model lacks inherent step-by-step reasoning strategies (arithmetic for Llama) but cannot compensate for deeply ingrained forward biases (logic for DeepSeek-R1).

4.7 Cross-Seed Stability

Table 3 shows that DRG measurements are stable across the three dataset seeds, with standard deviations below 2.6 pp in all cases.

4.8 Error Analysis

We analyze errors from the two models evaluated with CoT prompting (Llama-3.1-8B and DeepSeek-R1-7B) by categorizing backward-reasoning failures.

Propositional logic. The dominant backward error types are *rejecting the closed-world assumption* (models refuse to conclude "False" and instead state insufficient information) and *uncertainty hedging* (models qualify answers instead of committing). DeepSeek-R1-7B additionally shows *incorrect backward chaining* (23 of 50 sampled errors), where the model attempts backward reasoning but applies rules incorrectly.

Relational reasoning. The most common error is *wrong relationship type* (28–37 of 50 sampled errors across models), where models identify the correct entities but assign an incorrect relationship (e.g., responding "uncle" instead of "grandfather").

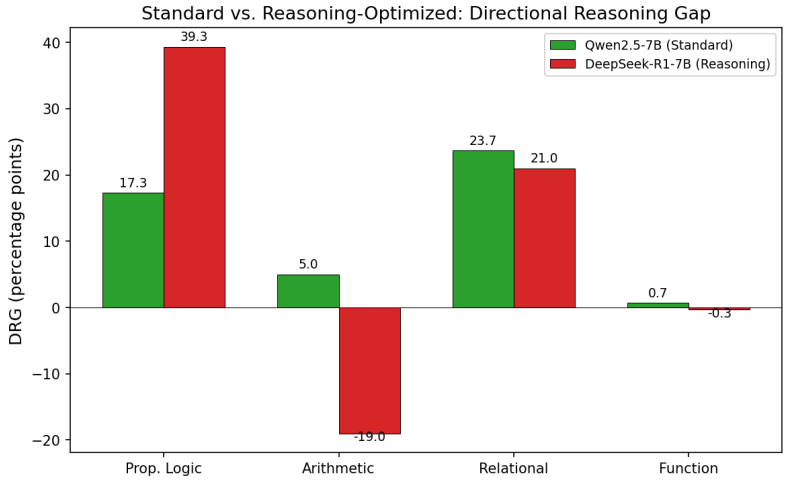


Figure 3: DRG comparison: standard (Qwen2.5-7B) vs. reasoning-optimized (DeepSeek-R1-7B) training across domains. Reasoning-optimized training reduces DRG in relational reasoning but amplifies it in propositional logic.

Table 2: Effect of chain-of-thought (CoT) prompting on DRG. Values are DRG in percentage points (seed 42). CoT substantially reduces the logic DRG for Llama but amplifies it for DeepSeek-R1.

Model	Prompt	Logic	Arithmetic	Relational	Function
Llama-3.1-8B	Standard	11.3	−14.0	24.7	−12.7
	CoT	2.0	0.3	23.0	−5.3
DeepSeek-R1-7B	Standard	39.3	−19.0	21.0	−0.3
	CoT	48.3	−1.7	18.7	0.0

Arithmetic. Errors split between *computation errors* on forward problems and *incorrect inverse operations* on backward problems. DeepSeek-R1-7B has a notably high parse failure rate (46 of 50 sampled arithmetic errors), indicating the model's chain-of-thought often fails to produce a clean final answer.

Function computation. Errors are rare for most models. Phi-3.5-mini is the exception, with 183 forward errors (out of 204 total) primarily due to computation mistakes when evaluating functions.

4.9 Overview Visualization

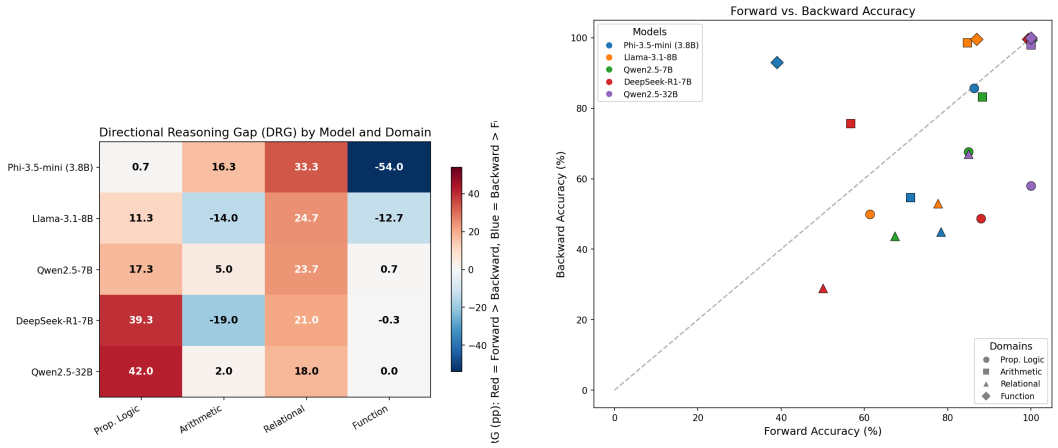
Figure 4a provides a summary heatmap of DRG values across all model-domain combinations, and Figure 4b shows the forward-versus-backward accuracy scatter plot.

5 Discussion

Directional asymmetry is real but not uniformly forward-biased. Our results partially confirm and partially refute the initial hypothesis. We find statistically significant directional asymmetry in propositional logic (5 models, 3 significant) and relational reasoning (all 5 significant), supporting the existence of systematic DRG. However, the hypothesis that the gap is always forward-biased is refuted: arithmetic and function computation domains often show backward advantages, particularly for Llama-3.1-8B and DeepSeek-R1-7B.

Table 3: Cross-seed stability of DRG (mean $\pm$ std across 3 seeds, in pp). All standard deviations are below 3 pp, confirming that FLIPBENCH produces reliable measurements.

Model	Prop. Logic	Arithmetic	Relational	Function
Llama-3.1-8B	9.7 ± 2.4	-14.7 ± 2.5	25.1 ± 1.9	-12.7 ± 0.8
DeepSeek-R1-7B	40.3 ± 1.7	-19.3 ± 0.3	24.2 ± 2.6	-0.2 ± 0.2



(a) DRG heatmap (models $\times$ domains). Red indicates forward bias; blue indicates backward advantage.

(b) Forward vs. backward accuracy. Points below the diagonal indicate forward bias. Points above indicate backward advantage.

Figure 4: Overview of directional reasoning asymmetry across all model-domain combinations.

Why do some domains show backward advantages? We hypothesize that the backward advantage in arithmetic and function computation for some models arises because the "backward" variants (find a missing operand; find an input given an output) may be better represented in instruction-tuning data as common word-problem formats. The "forward" variants (compute the result of a multi-step expression) may actually be harder for models that struggle with precise multi-step arithmetic—particularly at difficulty level 3, where Llama-3.1-8B's forward arithmetic accuracy drops to 54% while backward accuracy remains at 98%.

Reasoning-optimized training creates new asymmetries. DeepSeek-R1-7B's dramatically increased logic DRG (39.3 pp vs. Qwen2.5-7B's 17.3 pp) suggests that RL-based reasoning training may over-optimize for forward-chaining patterns, creating stronger directional biases in domains where forward reasoning templates are heavily rewarded during training.

Model scale partially addresses asymmetry. Qwen2.5-32B shows perfect forward accuracy in logic and arithmetic, but its backward logic accuracy (58.0%) lags far behind, producing the largest observed DRG (42.0 pp). Scale helps *absolute* performance but does not eliminate directional asymmetry—indeed, it may amplify it when forward performance saturates but backward performance does not.

Limitations. (1) Our benchmark covers four domains; generalization to other reasoning types (causal, spatial, temporal) is untested. (2) We evaluate five models; broader evaluation across more model families and sizes would strengthen conclusions. (3) The programmatic generation of problems, while ensuring correctness, may not capture the full complexity of natural-language reasoning tasks. (4) Some models (especially DeepSeek-R1-7B) have high parse failure rates, which may inflate

apparent error rates. (5) We use zero-shot evaluation; few-shot prompting might yield different patterns.

6 Conclusion

We introduced FLIPBENCH, a benchmark of 1,200 matched forward-backward reasoning pairs for measuring directional reasoning asymmetry in LLMs. Evaluating five models across four domains, we found that directional asymmetry is a real and domain-specific phenomenon: propositional logic and relational reasoning show robust forward biases (DRG up to 42.0 pp), while arithmetic and function computation can show backward advantages. The asymmetry is stable across dataset seeds (std < 2.6 pp) and amplifies with difficulty in relational reasoning ($p = 0.031$). Reasoning-optimized training does not uniformly reduce the gap, and chain-of-thought prompting has mixed domain-dependent effects.

These findings establish directional consistency as a diagnostic dimension complementary to aggregate accuracy. We believe FLIPBENCH can serve as a tool for diagnosing whether models exhibit genuine bidirectional reasoning ability—a property critical for applications requiring backward inference such as diagnosis, debugging, and root cause analysis. Future work should extend the benchmark to additional reasoning domains, evaluate larger model families, and investigate training-time interventions to reduce directional asymmetry.

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A Additional Results

A.1 Forward vs. Backward Accuracy by Model

Figure 5 provides a detailed accuracy breakdown.

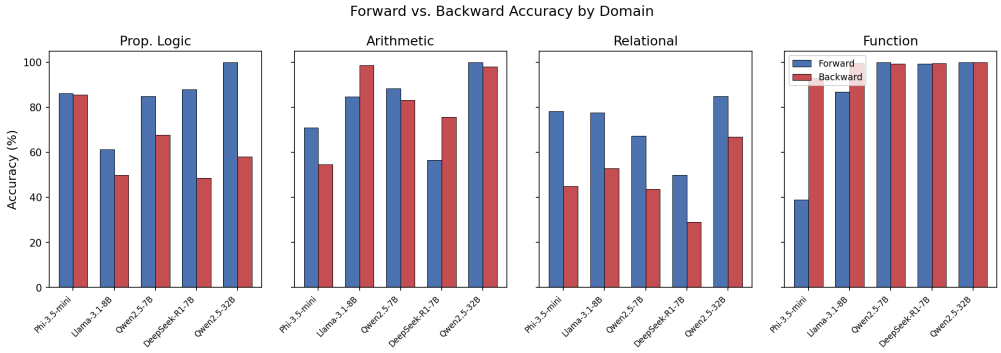

Figure 5: Forward and backward accuracy breakdown per model and domain.

A.2 Chain-of-Thought Ablation Visualization

Figure 6 shows the CoT ablation results visually.

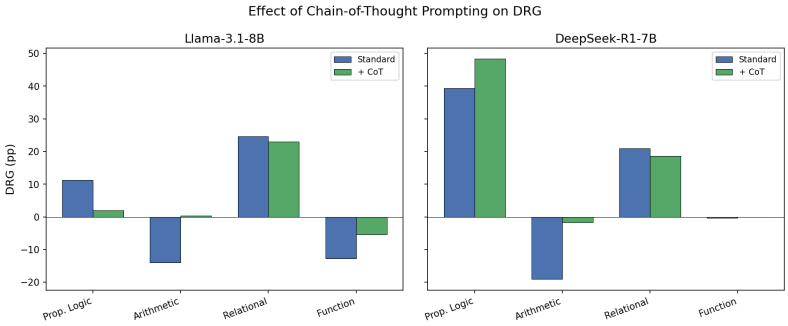

Figure 6: DRG with and without chain-of-thought prompting for Llama-3.1-8B and DeepSeek-R1-7B across four domains.

A.3 Statistical Test Details

Table 4 reports detailed McNemar's test statistics for all model-domain combinations.

Table 4: McNemar's test results for directional asymmetry. p-values are Bonferroni-corrected across 20 tests (5 models $\times$ 4 domains).

Model	Statistic	Prop. Logic	Arithmetic	Relational	Function
Phi-3.5-mini	DRG (pp)	0.7	16.3	33.3	-54.0
	p (corrected)	1.000	<0.001	<0.001	1.000
Llama-3.1-8B	DRG (pp)	11.3	-14.0	24.7	-12.7
	p (corrected)	0.176	1.000	<0.001	1.000
Qwen2.5-7B	DRG (pp)	17.3	5.0	23.7	0.7
	p (corrected)	<0.001	0.357	<0.001	1.000
DeepSeek-R1-7B	DRG (pp)	39.3	-19.0	21.0	-0.3
	p (corrected)	<0.001	1.000	<0.001	1.000
Qwen2.5-32B	DRG (pp)	42.0	2.0	18.0	0.0
	p (corrected)	<0.001	0.313	<0.001	1.000